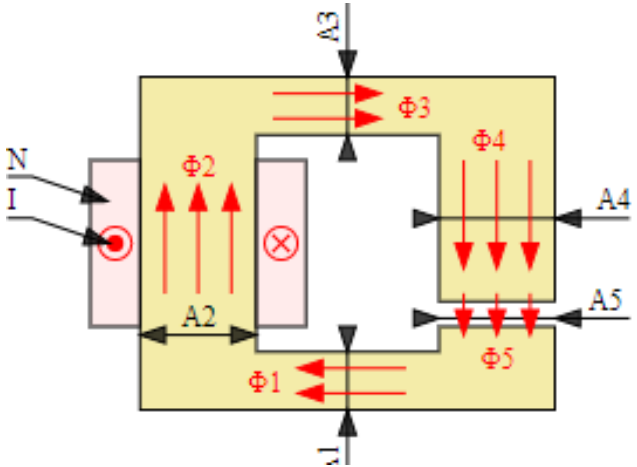
ECE 330

Chapter 4

Magnetic circuits, Mutual Inductance, and Transformers

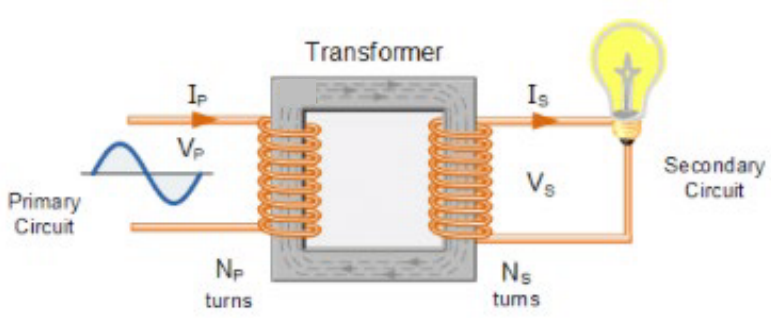


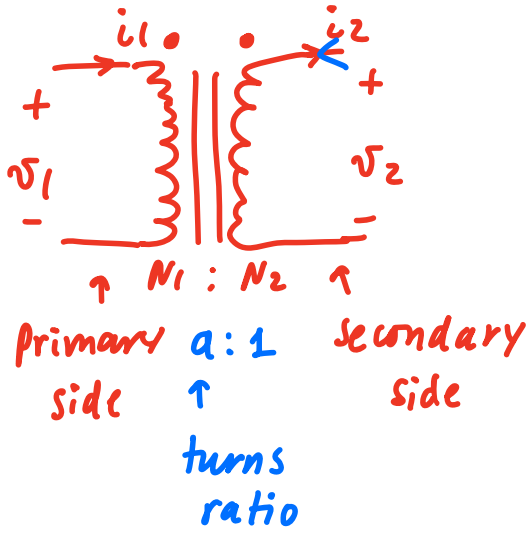
Prof.Olga Mironenko
University of Illinois at Urbana-Champaign
Fall 2026

Learning Objectives: you will be able to

- Analyze **ideal transformers,** and determine voltage, current and impedance relationships between primary and secondary sides.
- Use **Maxwell's equations** to analyze an inductor, and to develop **magnetic equivalent circuits.**
- Model magnetic systems using **magnetic equivalent circuits**, including **reluctance** and **inductance**.
- Evaluate airgap effects, accounting for **airgap reluctance** and **fringing effects.**
- Analyzed **coupled coils**, using **mutual inductance** and **the dot convention.**

Ideal Transformers





$$\frac{v_1}{v_2} = \frac{N_1}{N_2} = a$$

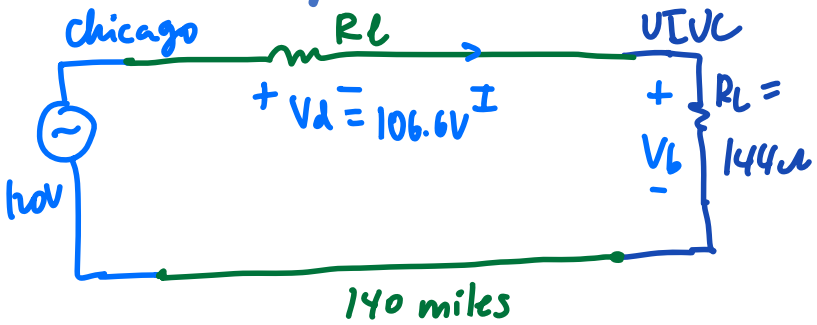
$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

Q: p_1 and p_2 -?

- $p_1 = v_1 i_1 \quad \frac{N_1}{N_2} \cdot i_1 = a \cdot i_1$
- $p_2 = v_2 i_2 = v_1 i_1 = p_1 \Rightarrow$ Power across ideal transformer is the same.

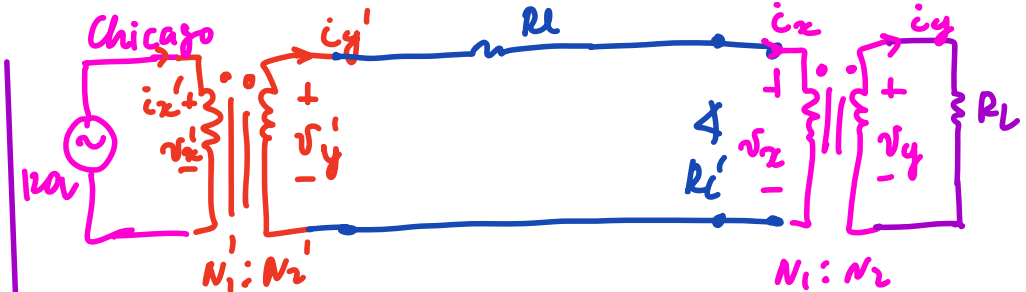
$\uparrow \frac{N_2}{N_1} v_1 = \frac{1}{a} \cdot v_1$

Why Edison lost the War of Currents



$$* R_l = \rho \frac{l}{S}$$

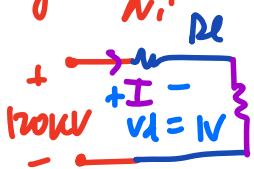
R_l, Ω	I	V_l	η
1148Ω	93 mA	13.4V	11.2%



1 : 1000
step-up transformer

1000 : 1
step-down transformer

$$\bullet\ v_y' = \frac{N_2'}{N_1'} \cdot v_x' = 120kV$$



$$R_l' = (1000)^2 \cdot 144 = 144M\Omega$$

$$I = \frac{120kV}{144M + 1148} = 0.83mA$$

$$\eta = 99.9\%$$

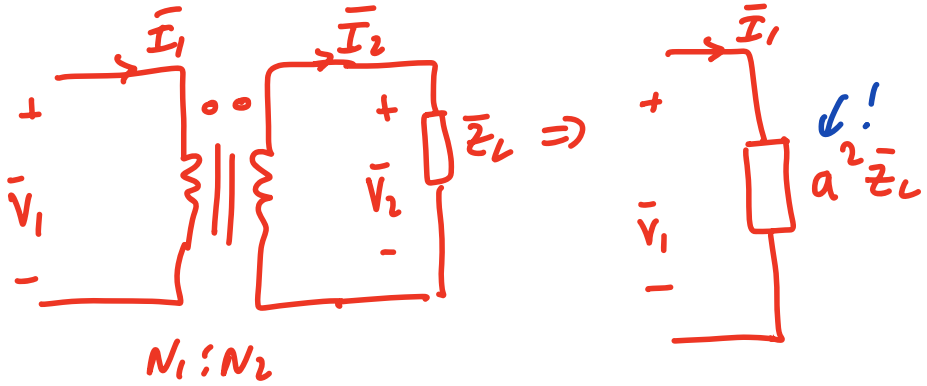
Move impedance across the transformer

$$\bullet\ R_L = \frac{v_y}{i_y} = \frac{N_2 \cdot v_x}{N_1 \left(\frac{N_1}{N_2}\right) i_x}$$

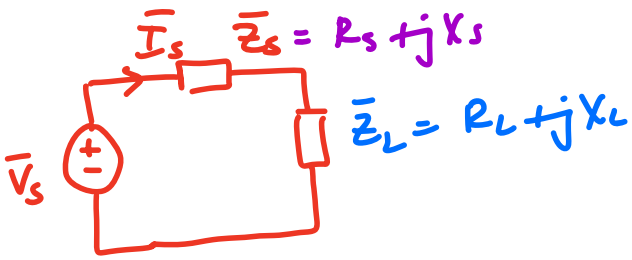
$$\bullet\ \underbrace{\frac{v_x}{i_x}}_{R_l'} = \left(\frac{N_1}{N_2}\right)^2 \cdot R_L = a^2 \cdot R_L$$

$$\boxed{\bar{Z}_p = a^2 \cdot \bar{Z}_s}$$

Moving Impedances Across a Transformer



Maximum Power Transfer



Q: What $\bar{Z}_L$ to choose for max. power transfer?

- $P_L = |\bar{I}_s|^2 \cdot R_L = \dfrac{|\bar{V}_s|^2 \cdot R_L}{(R_s+R_L)^2 + (X_s+X_L)^2}$
- $\bar{I}_s = \dfrac{\bar{V}_s}{\bar{Z}_s+\bar{Z}_L}$

$\Downarrow$ $\quad$ $\Downarrow$

$X_L = -X_s$

$R_L = R_s$

$\Downarrow$

- $X_L = -X_s$
- $R_L = R_s$

$\Downarrow$

$\bar{Z}_L = \bar{Z}_s^*$

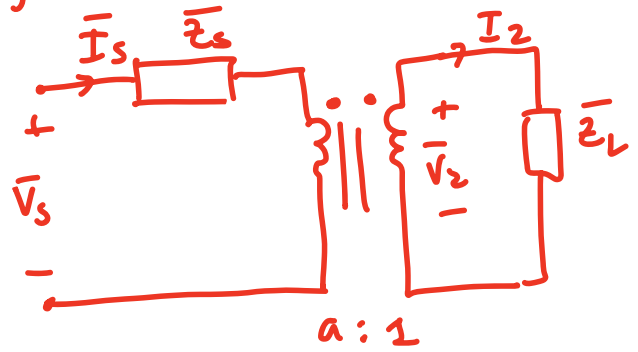
Maximum Power Transfer (Continued)

- Can show for maximum power dissipated at the load
  - The load impedance is the complex conjugate of the line impedance ($\bar{Z}_L = \bar{Z}_s^*$)
  - For a fixed ratio between the resistive and reactive terms of the load impedance, the magnitude of the load impedance is the same as the line impedance

$$|\bar{Z}_L| = |\bar{Z}_s| \text{ for } \frac{R_L}{X_L} = c \text{ (fixed as constant)}$$

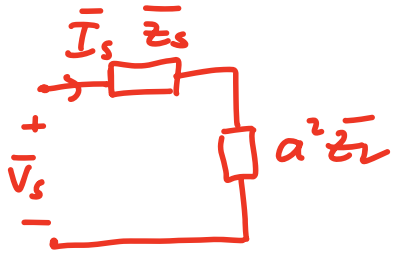
Maximum Power Transfer with Transformer

* Imposing $|\bar{Z}_L| = |\bar{Z}_s|$ can be done with a transformer:



$\Rightarrow$

Moving impedance across the transformer first:

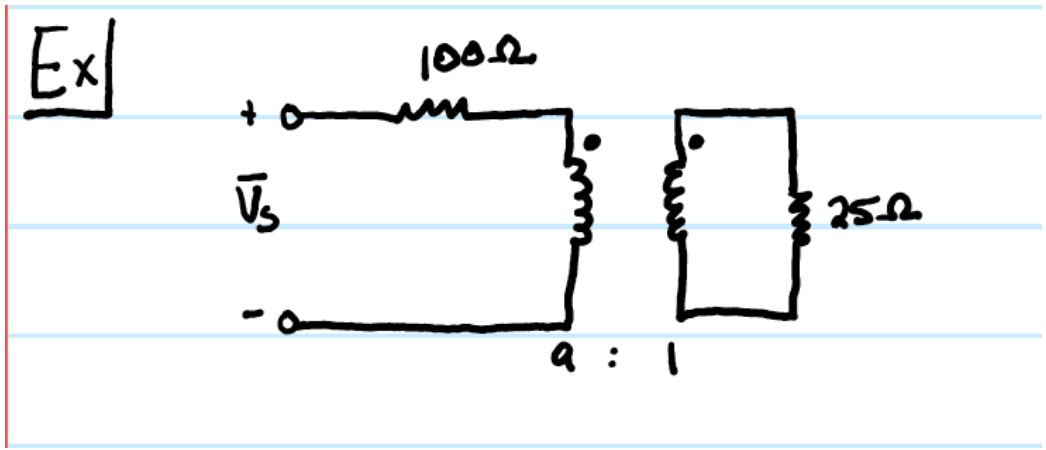


$\Downarrow$

$$|\bar{Z}_s| = |a^2 \bar{Z}_L|$$

can pick "a" for $a = \sqrt{\frac{|\bar{Z}_s|}{|\bar{Z}_L|}}$ a max. power transfer

Example 1



What is *a* for maximum power transfer?

a) 1

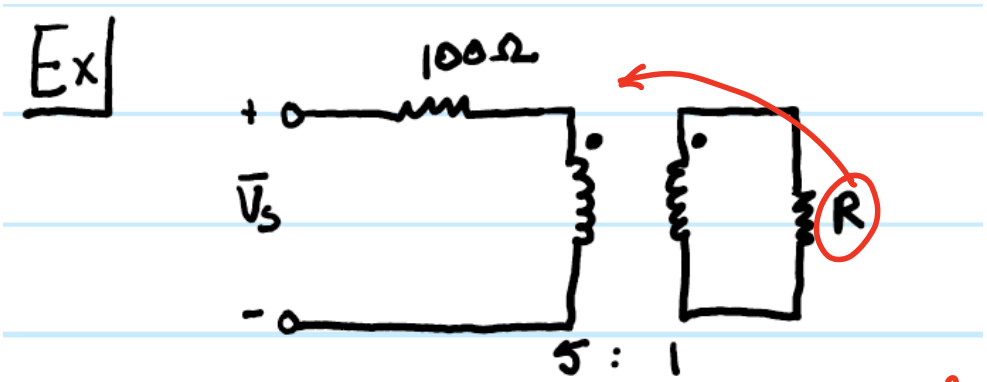
b) 2

c) 3

d) 4

$$a = \sqrt{\frac{R_S}{R_L}} = \sqrt{\frac{100}{25}} = 2$$

Example 2

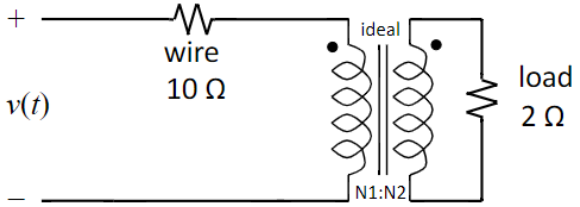


$100 = 5^2 \cdot R \Rightarrow R = 4\Omega$

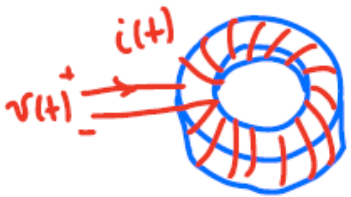
What is *R* for maximum power transfer?

Example 3 (Pr 4d, Spring 2021, M1)

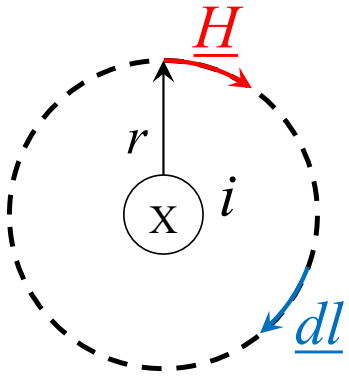
d) The purpose of our transformer is to drive a 2 Ω load from a source that is connected through a 10 Ω wire, as shown below. If v(t) = cos(t) V and we model the transformer as ideal, what is the average power delivered to the load?



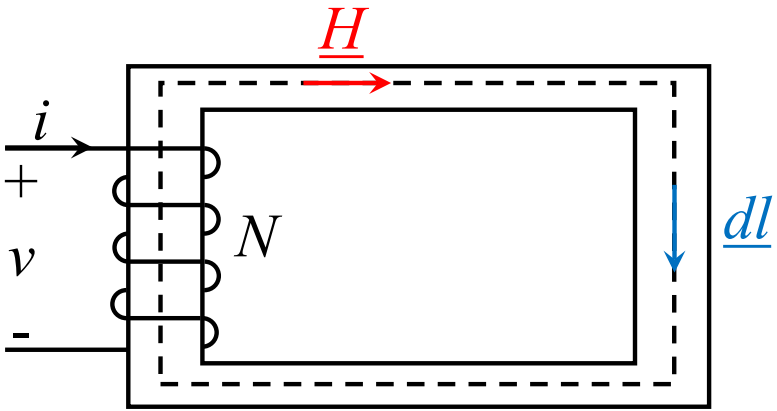
Electromagnetics



Electromagnetics



Electromagnetics



Magnetic Equivalent Circuit

Magnetic **Electrical**

What about the *E*?

Inductance

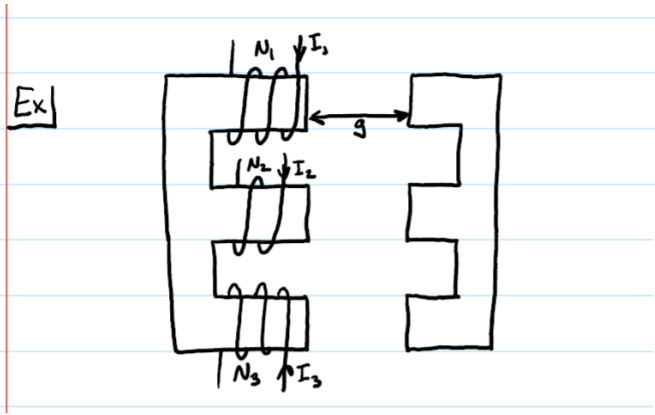
Iron Core with Air Gap

Iron Core with Air Gap

Faraday's Law

Fringing Effects

Example 4



Given: $N_1 = 100$ $I_1 = 25A$ $A_g = 4cm^2$

$N_2 = 50$ $I_2 = 10A$ $\mu = \infty$

$N_3 = 100$ $I_3 = 15A$ $g = 0.1cm$

Where should the positive terminals be placed on the MMF sources?

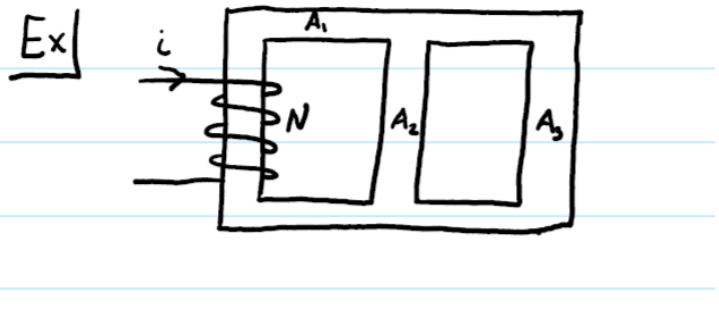
a) right, left, right

b) left, left, right

c) right, right, left

d) right, right, right

Example 5



Coupled Coils: ideal case

Dot convention

Dot convention

- Step 1) Select a coil and put a dot on one terminal- arbitrarily (your choice)
- Step 2) Mark the dotted terminal with + voltage polarity and inject current into first dot
- Step 3) Use RHR to determine flux direction
- Step 4) Determine current direction in the second winding which will throw flux in the same direction as the first winding
- Step 5) Put the dot in the second winding, where current enters the terminal
- Step 6) Put + voltage polarity on the second dotted terminal

Coupled Coils: practical case

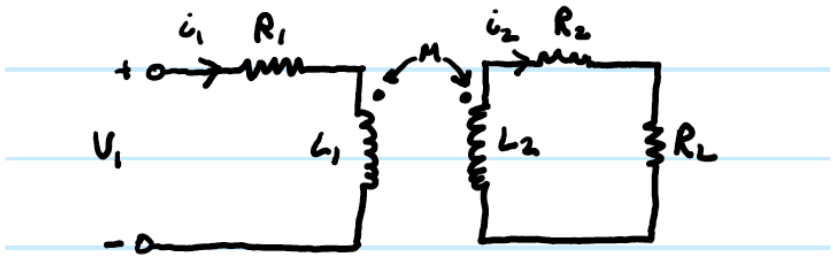
Ideal:

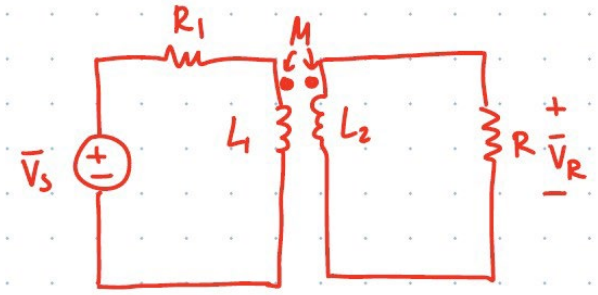
Practical:

Coefficient of Coupling

Example 6

Real Transformers





Given: ω, $\bar{V}_S$, R_1, R, L_1, L_2, M
Find: $\bar{V}_R$

- Real transformers have resistance in the wire, inductance from the core, and mutual inductance between the two windings.

Side note: Thevenin Equivalent (not on quizzes/exams)

